

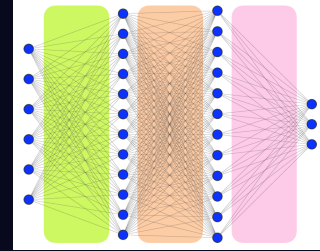
Lecture 2: Generalization and Regularization

Dr Tiffany Vlaar
University of Glasgow

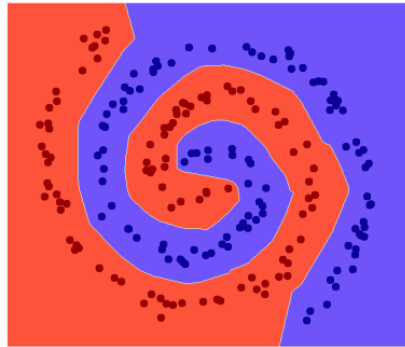
Try to enhance our understanding of neural networks

And use these findings to reduce computational cost

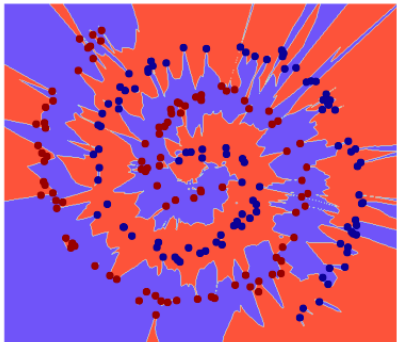
while enhancing or maintaining generalization.



Generalization



(a) 100% train, 100% test



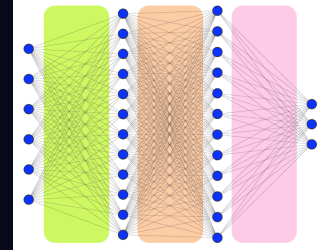
(b) 100% train, 7% test

Huang et al., 2020

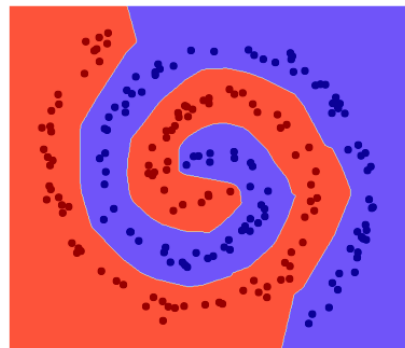
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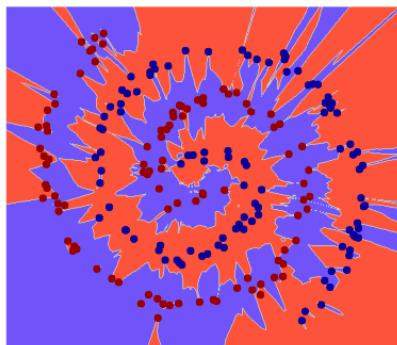
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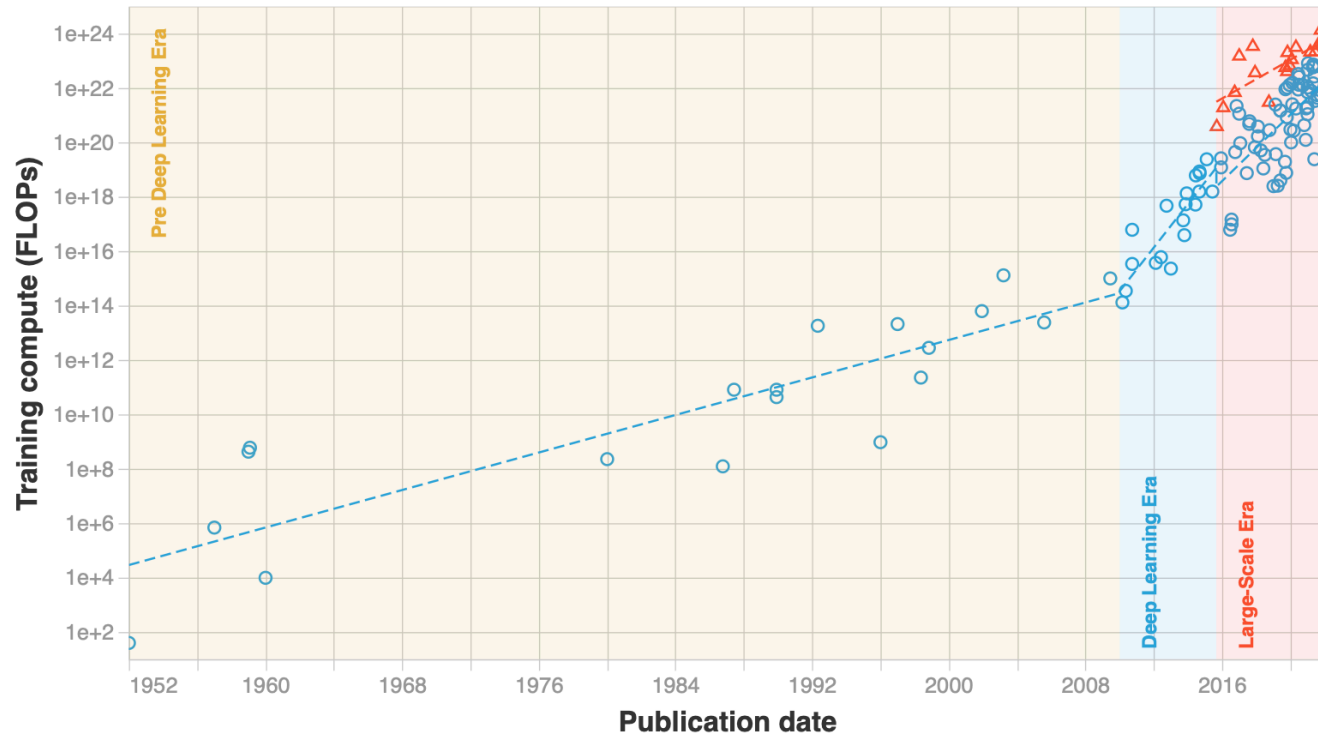
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Computational Cost

Training compute (FLOPs) of milestone Machine Learning systems over time

n = 121

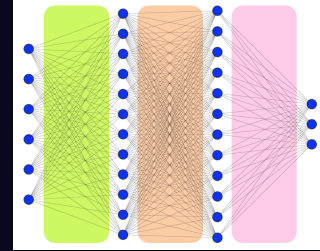


Sevilla et al., 2022

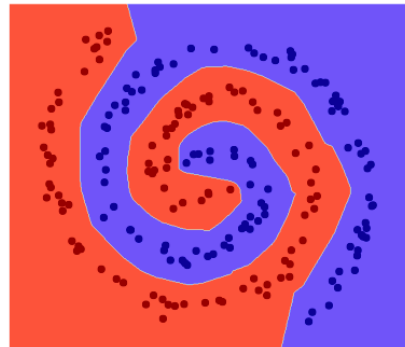
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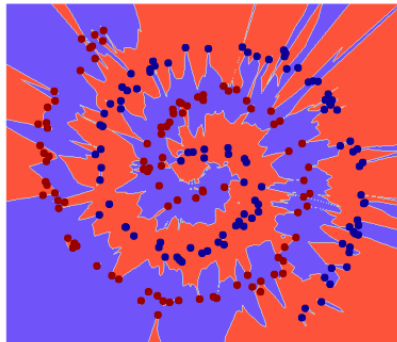
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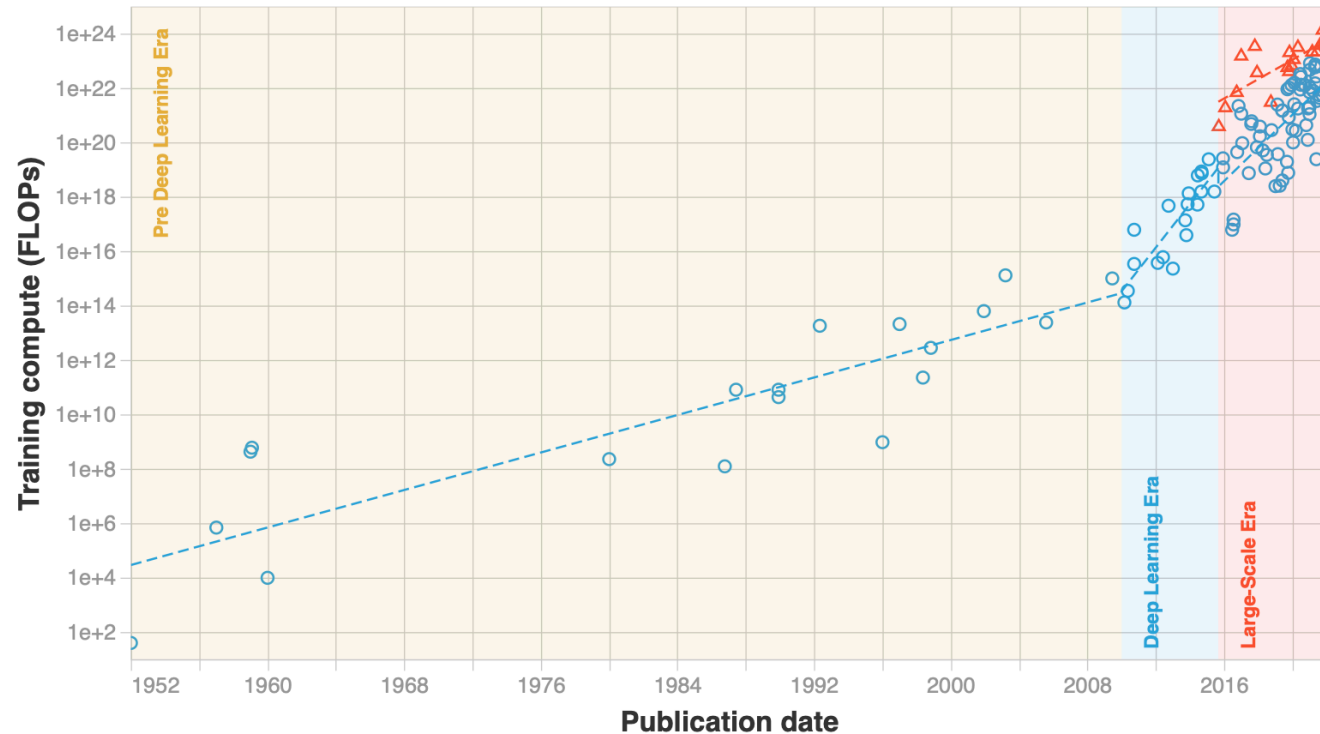
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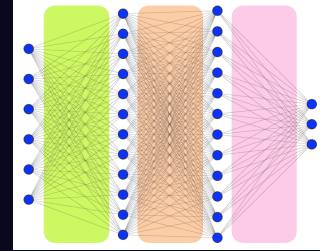
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Are these metrics sufficient.... future lecture

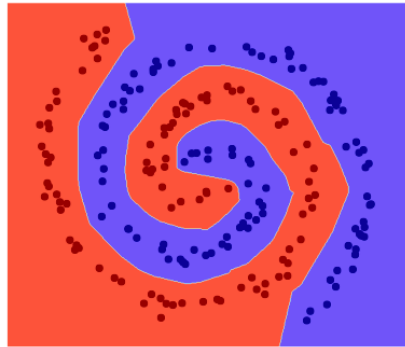
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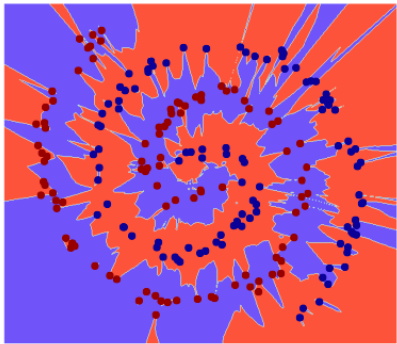
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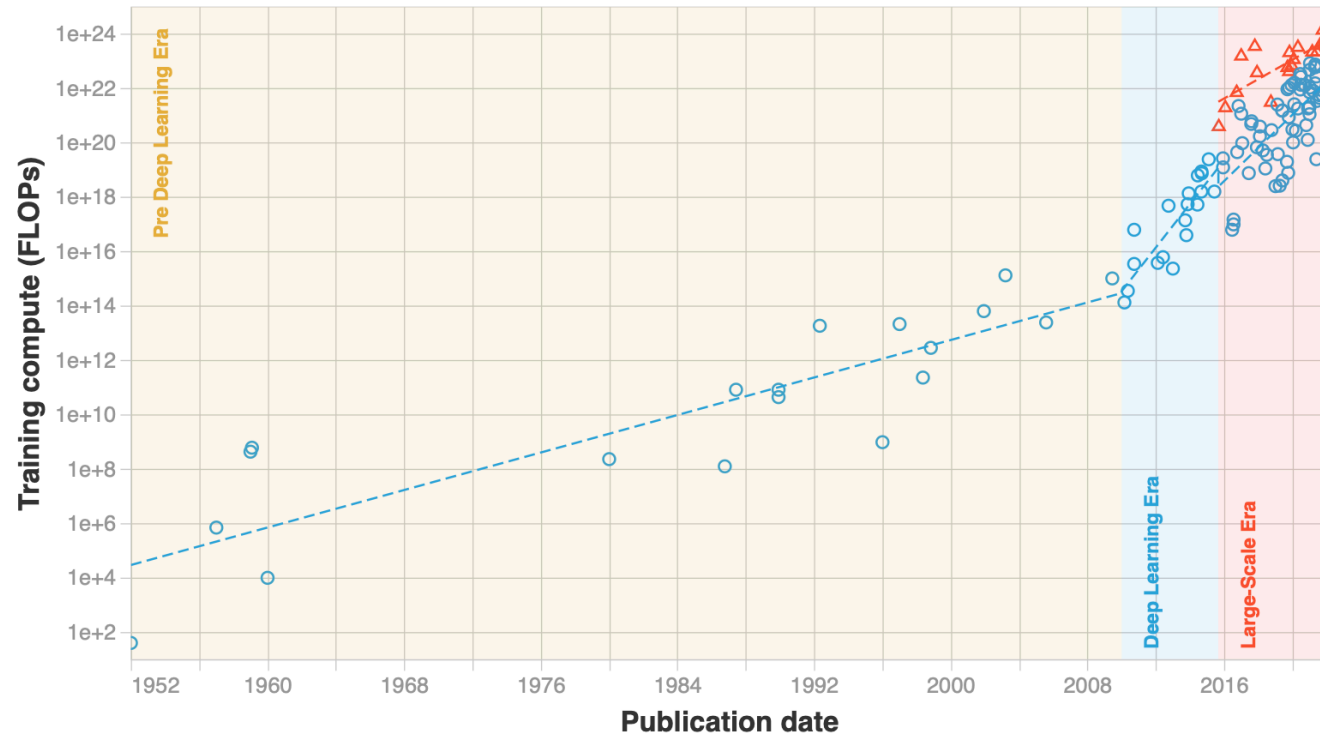


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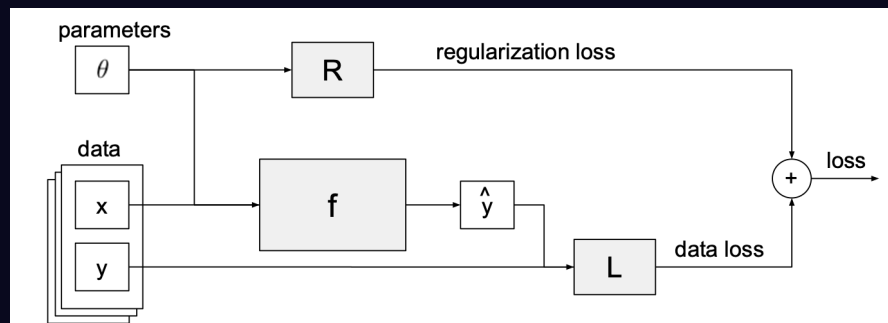
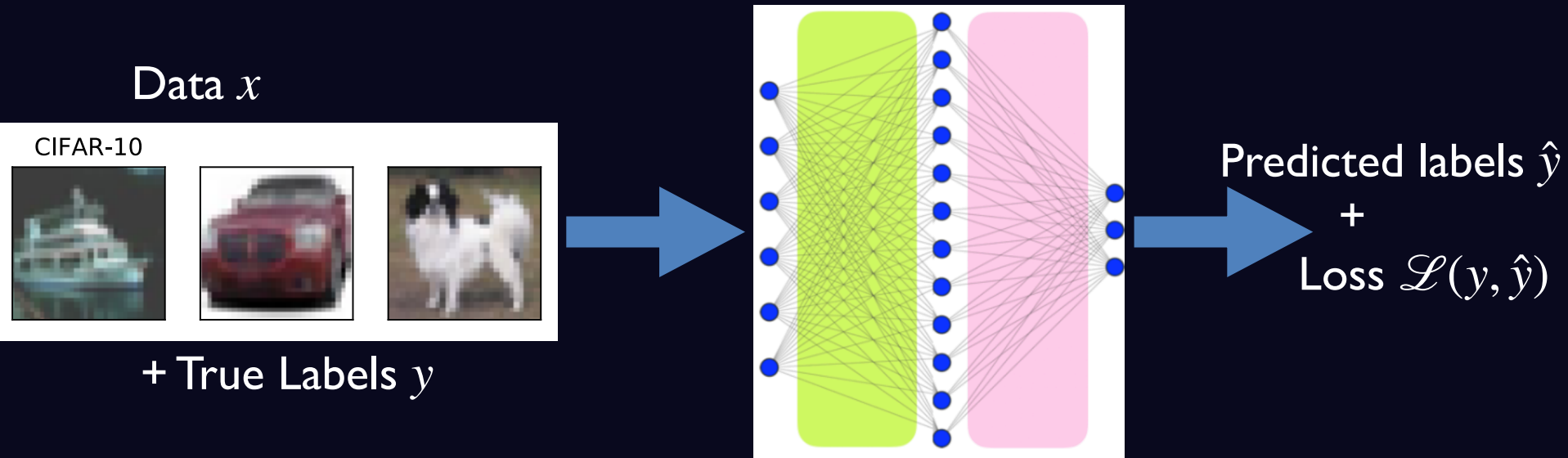
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Sevilla et al., 2022

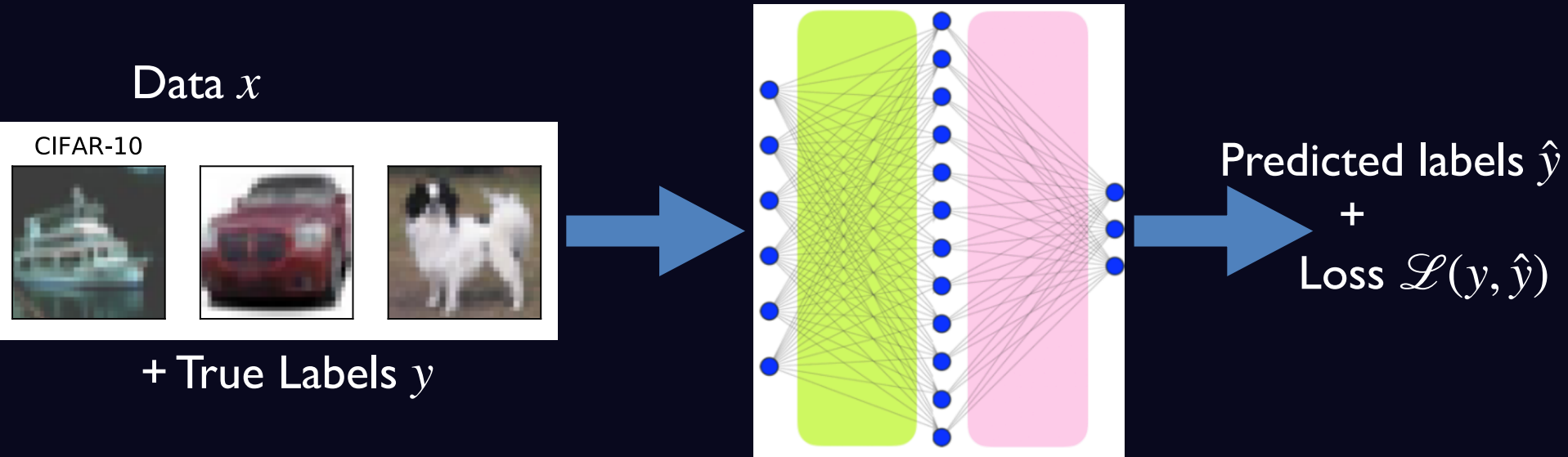
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Framework

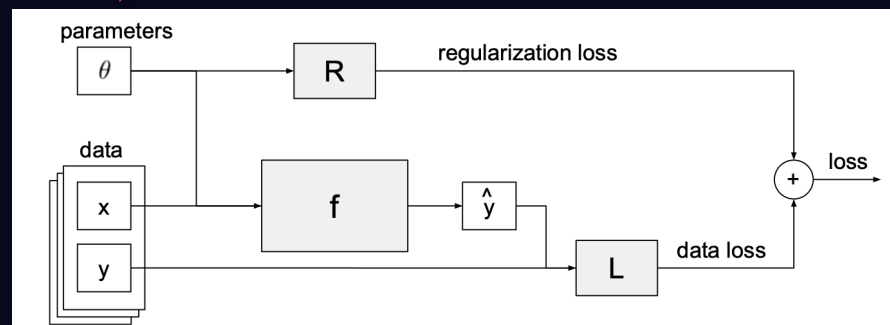


Karpathy, 2016

Framework

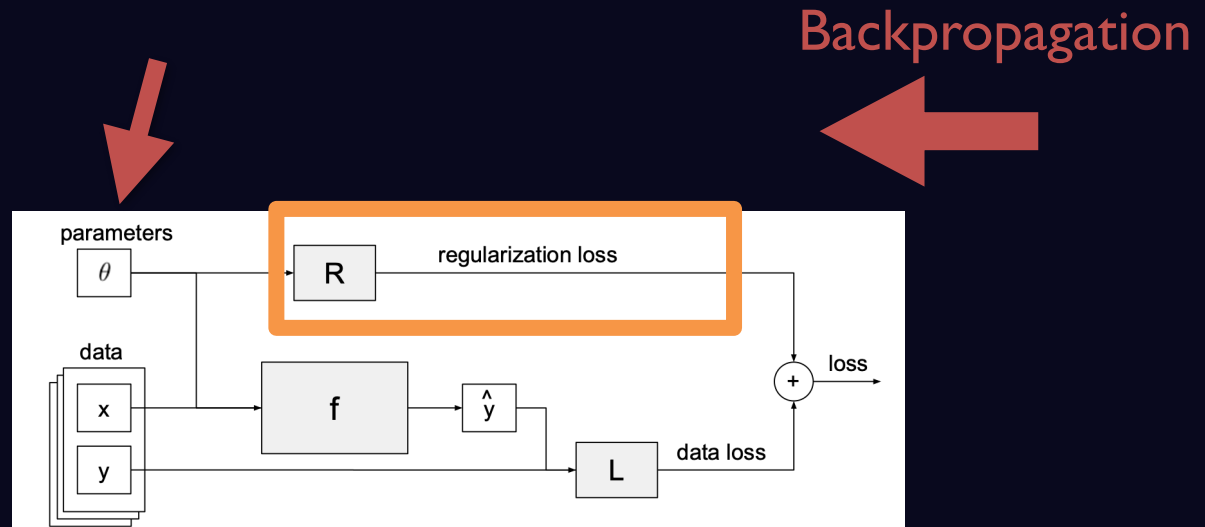
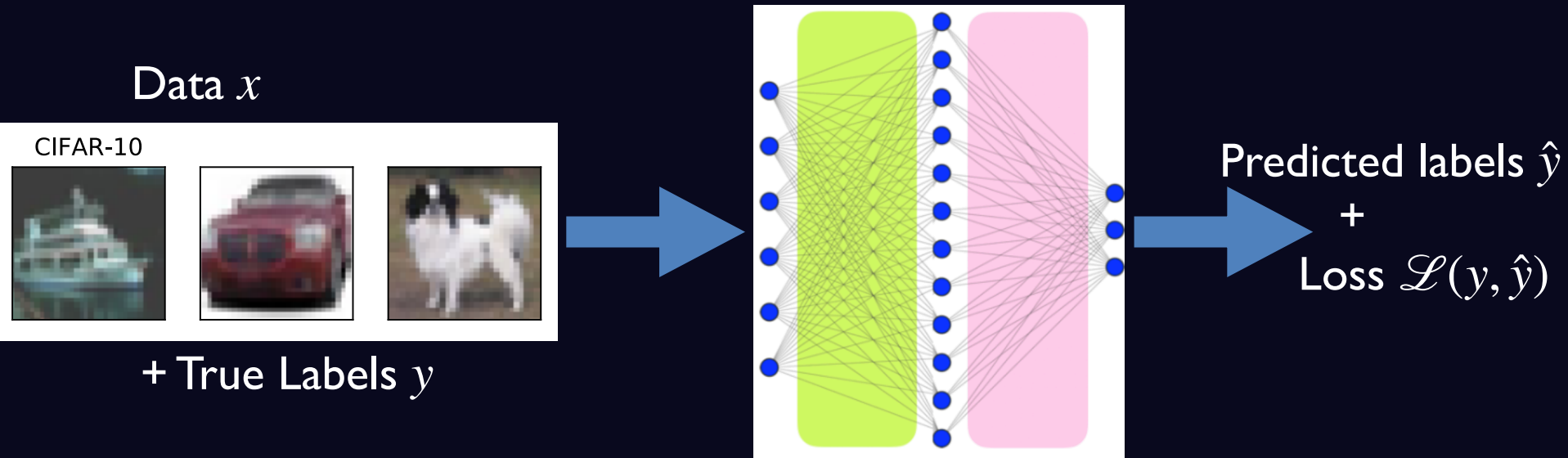


Backpropagation



Karpathy, 2016

Framework



Karpathy, 2016

Polynomial regression

Find best polynomial fit for a set of datapoints.

For example find w_0, w_1, w_2, w_3

$$y = w_3x^3 + w_2x^2 + w_1x + w_0$$

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Trade-off: Low degree polynomial does not fit data well enough, whereas high degree polynomial will have tendency to overfit.

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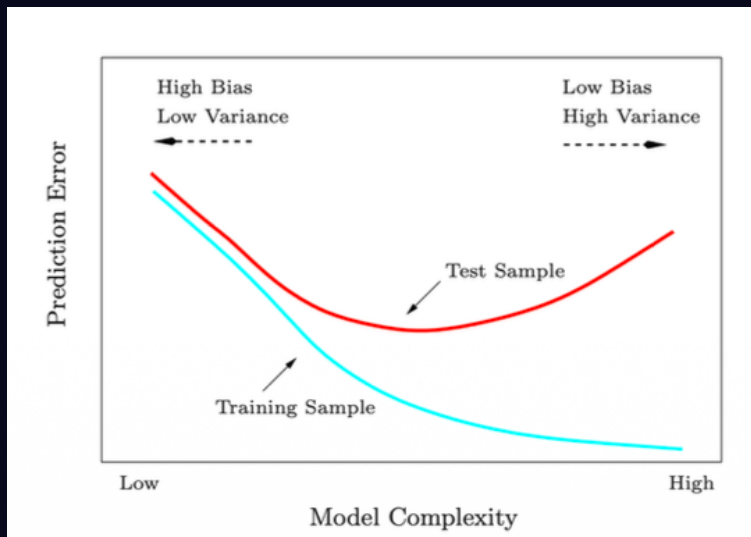
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What is the problem with overfitting?

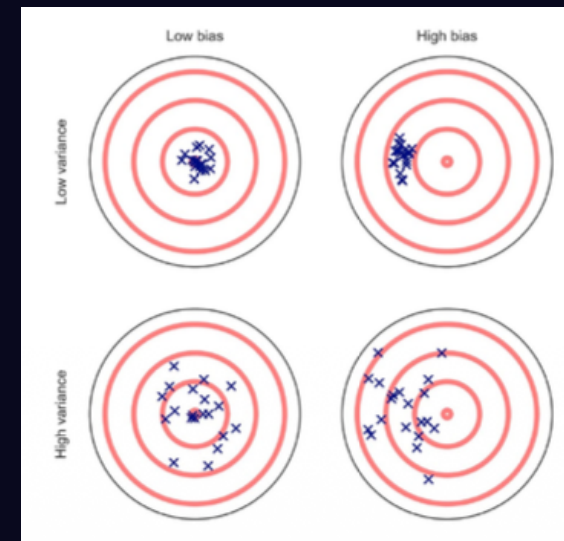
⇒ Perform well on provided data, but poorly on unseen data.

Bias-Variance Tradeoff

Trade-off: Low degree polynomial does not fit data well enough, whereas high degree polynomial will have tendency to overfit.



Hastie et al., 2017



Yarkoni and Westfall, 2017

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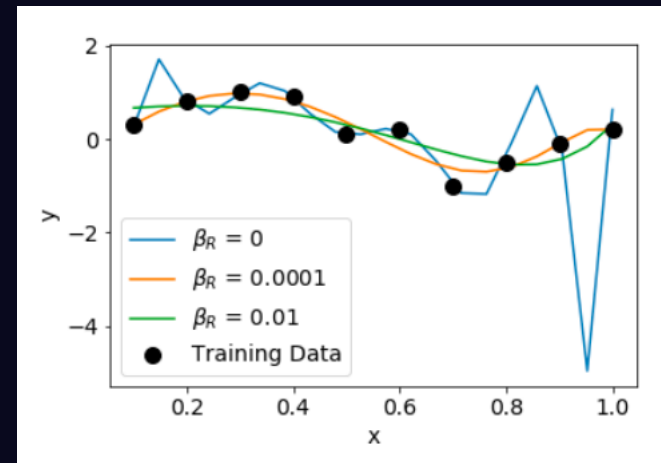
⇒ Perform well on provided data, but poorly on unseen data.

⇒ Mediate this using regularization.

Regularization strength β_R

Solving for 10th order polynomial

$$w = (x^T x + \beta_R I)^{-1} x^T y$$



Polynomial regression

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Example with soft constraints:

$$y = w_3x^3 + w_2x^2 + w_1x + w_0$$

such that $\sum_{i=0}^3 w_i^2 \leq k$.

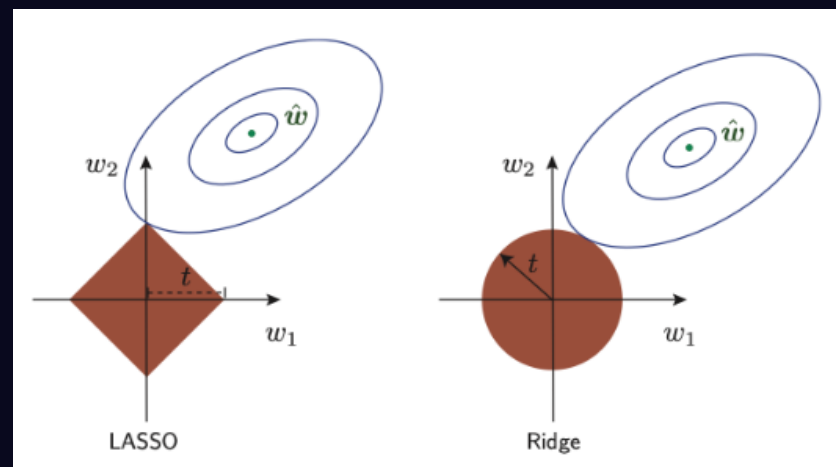
Regularized regression

1. Lasso regression [Tibshirani, 1996]

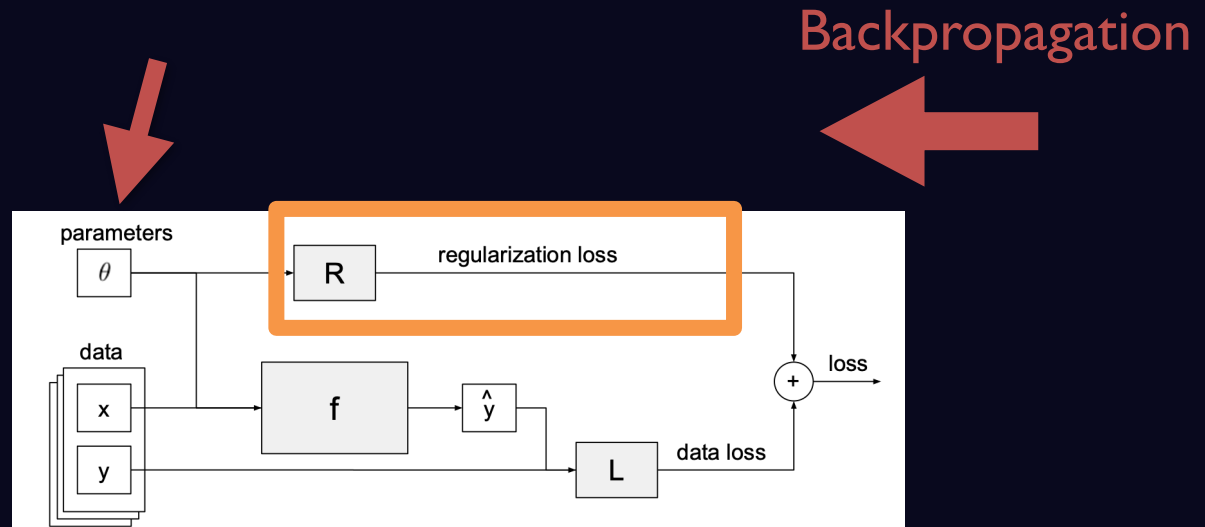
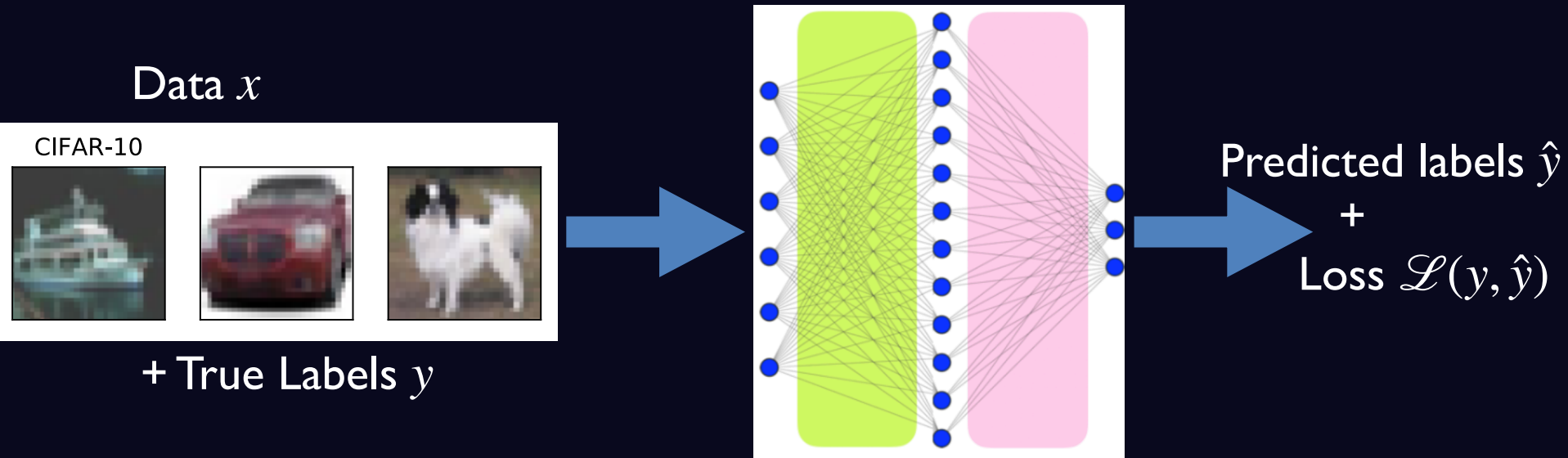
- L_1 regularization
- Many applications, but less used in machine learning
- Induces sparsity

2. Ridge regression [Hoerl and Kennard, 1970]

- L_2 regularization
- Used in machine learning
- Parameter shrinkage



Framework



Karpathy, 2016

Regularization

Training schemes in ML, for example:

Stochastic Gradient Descent (SGD)

+ often L_2 Regularization / weight decay

Adds an regularization term with strength β_R to the loss that penalizes the size of the norm of the weights:

$$\text{Loss} + \beta_R R(W)$$

Additional hyperparameter to tune: the regularization strength.

$$\text{For } L_2 \text{ Regularization: } \text{Loss} + \frac{\beta_R}{2} \|W\|_2^2$$

Why use Constraints for Neural Network Training?

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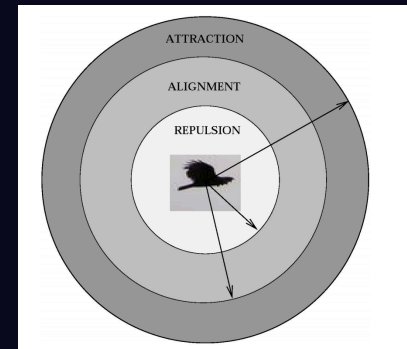
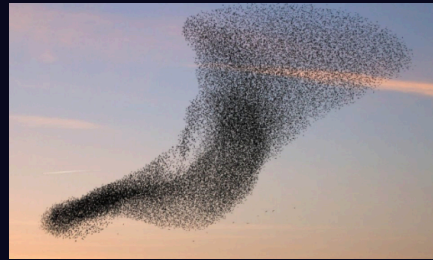
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Why use Constraints for Neural Network Training?

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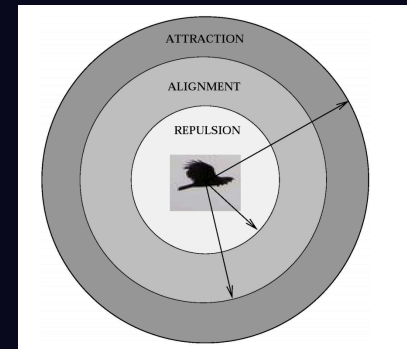
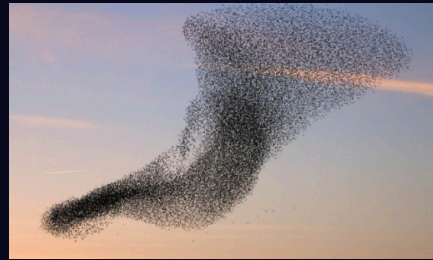
- Imposing good priors improves performance
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 - ⇒ Graph Neural Networks for modelling collective dynamics [Gaskell et al., 2024].



Cañizo et al. 2010

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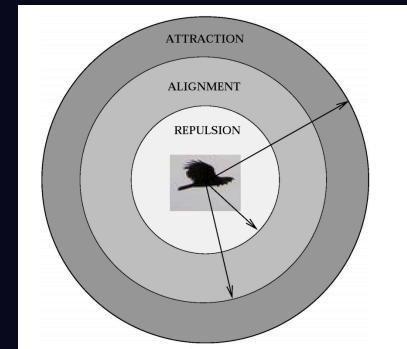
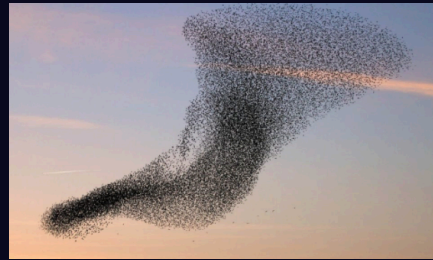


Cañizo et al. 2010

- Direct control of parameter space.

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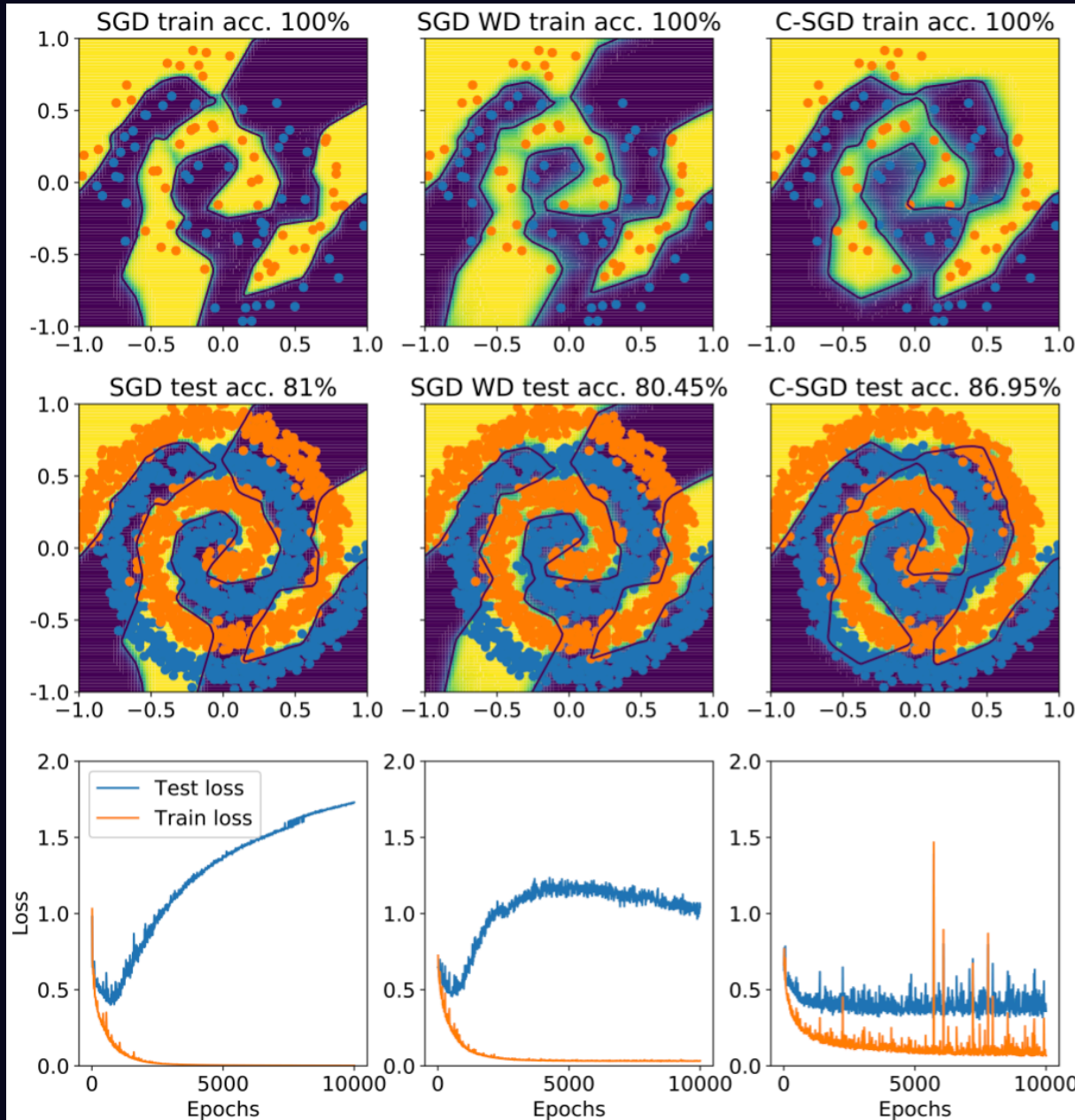
Cañizo et al. 2010

- Direct control of parameter space.
 - ⇒ Soft constraints vs. hard constraints
 - ⇒ General framework to impose hard constraints throughout training [with Leimkuhler, Pouchon, and Storkey].

Unconstrained

Soft Constraint

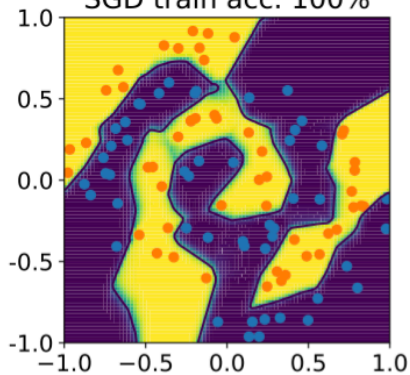
Hard Constraint



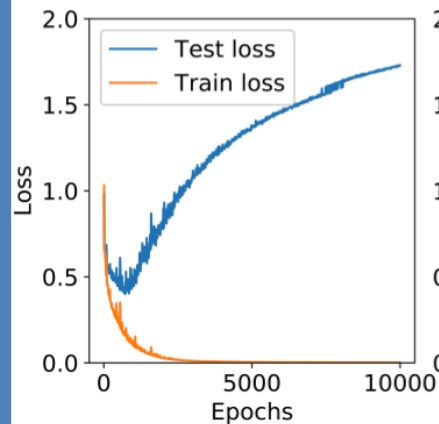
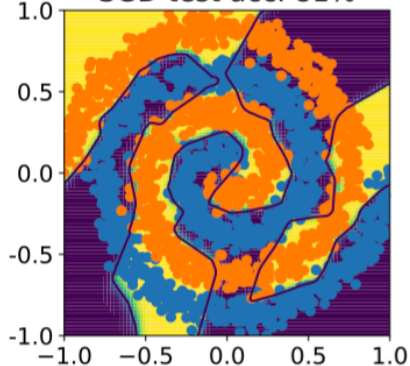
Magnitude-based Constraints

Unconstrained

SGD train acc. 100%

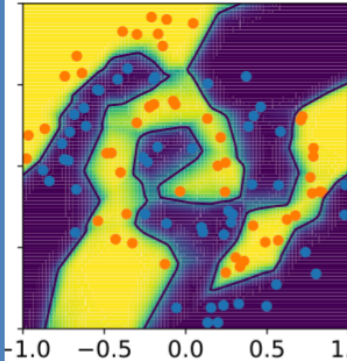


SGD test acc. 81%

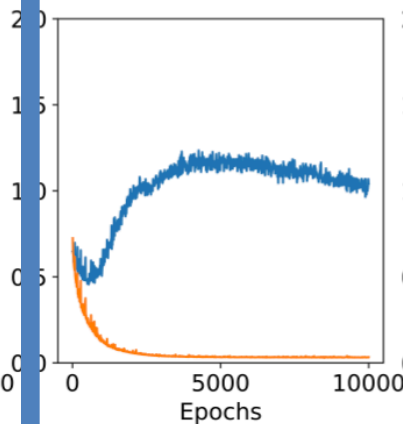
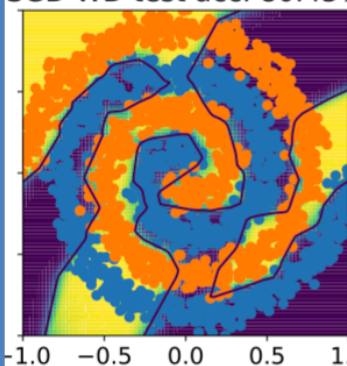


Soft Constraint

SGD WD train acc. 100%

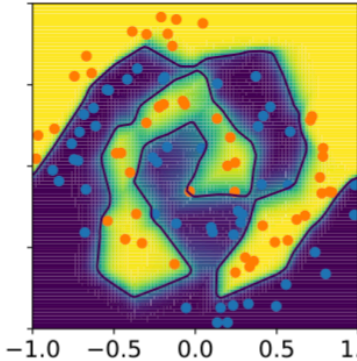


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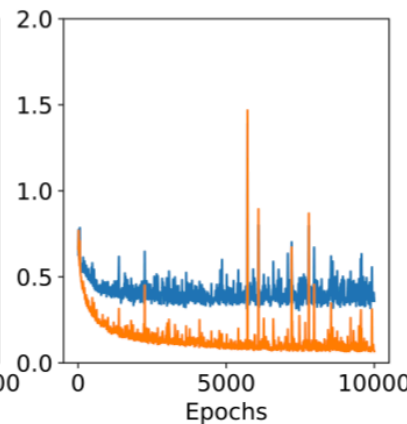
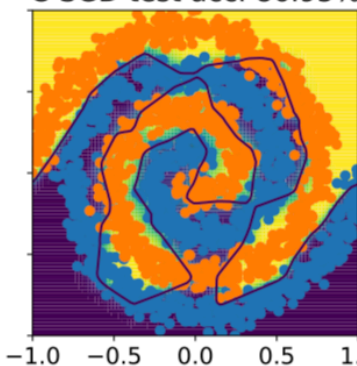


Hard Constraint

C-SGD train acc. 100%

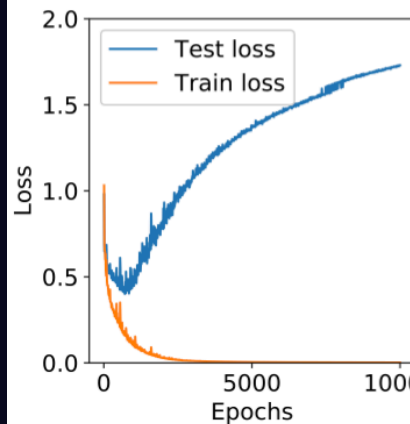
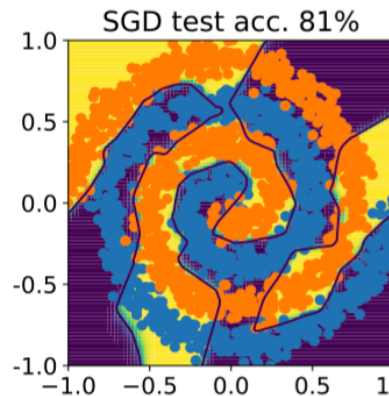
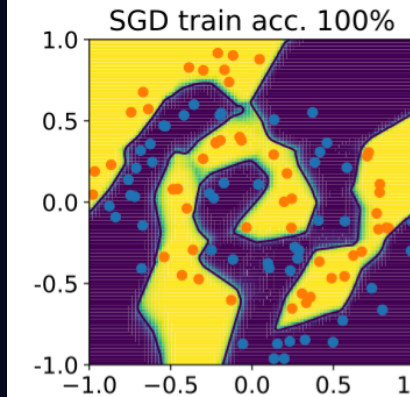


C-SGD test acc. 86.95%

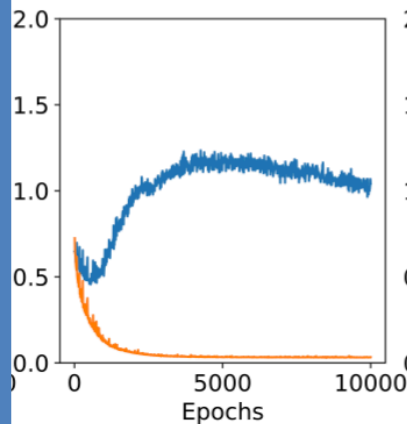
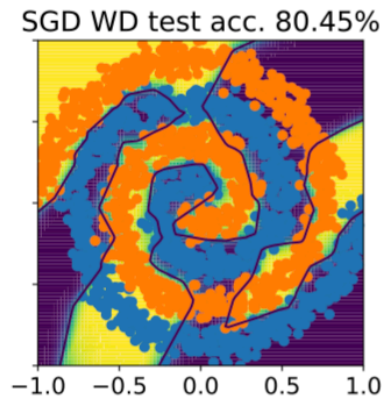
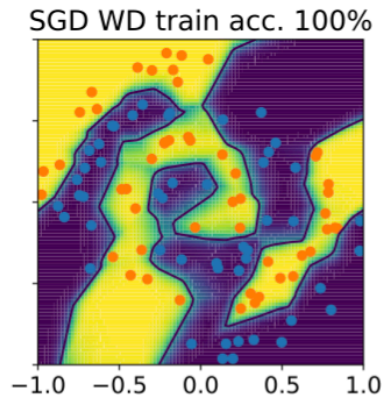


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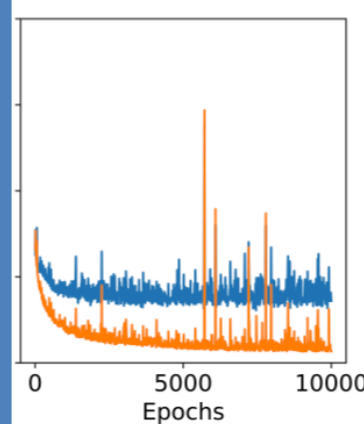
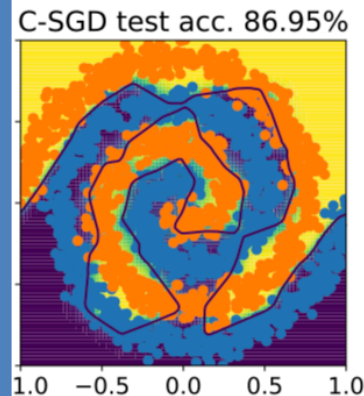
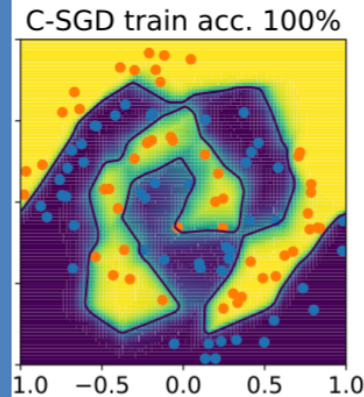
Unconstrained



Soft Constraint



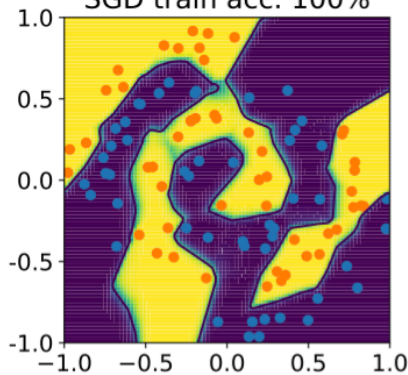
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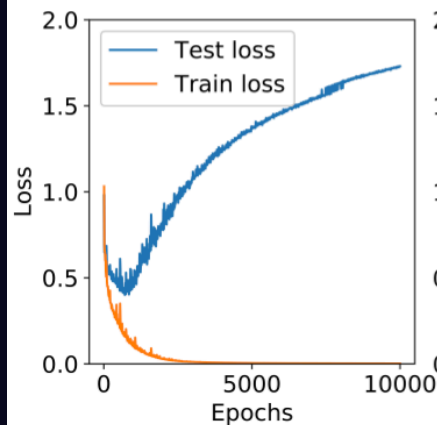
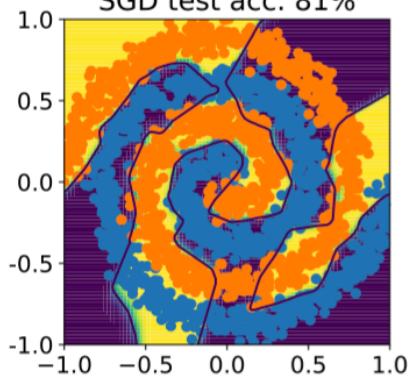
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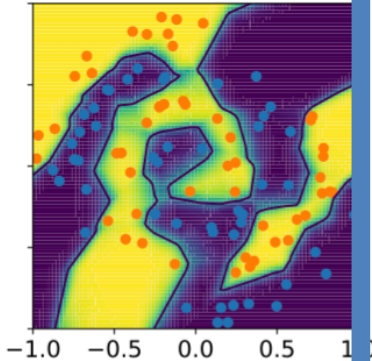


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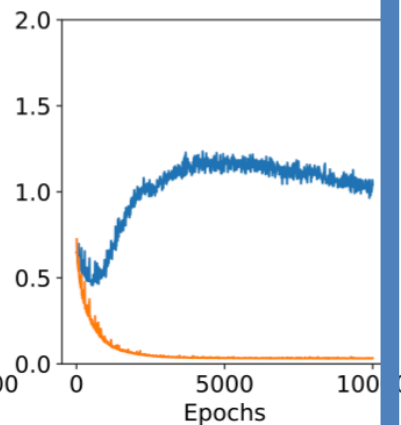
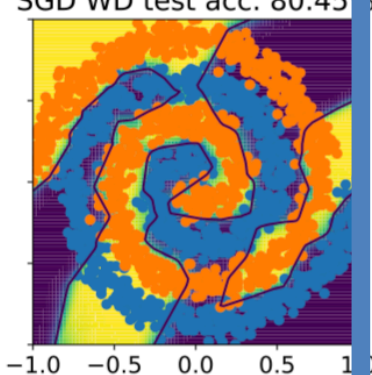


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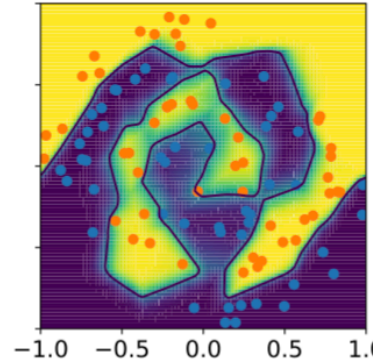


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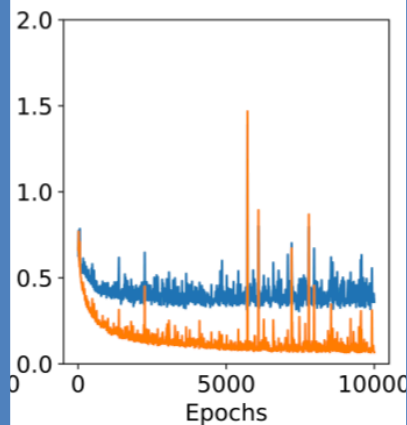
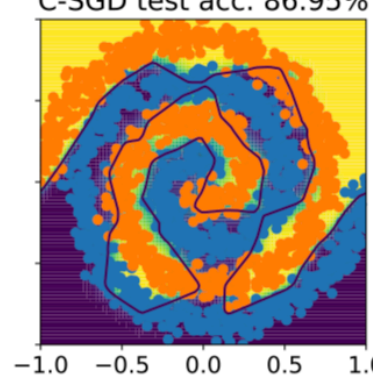


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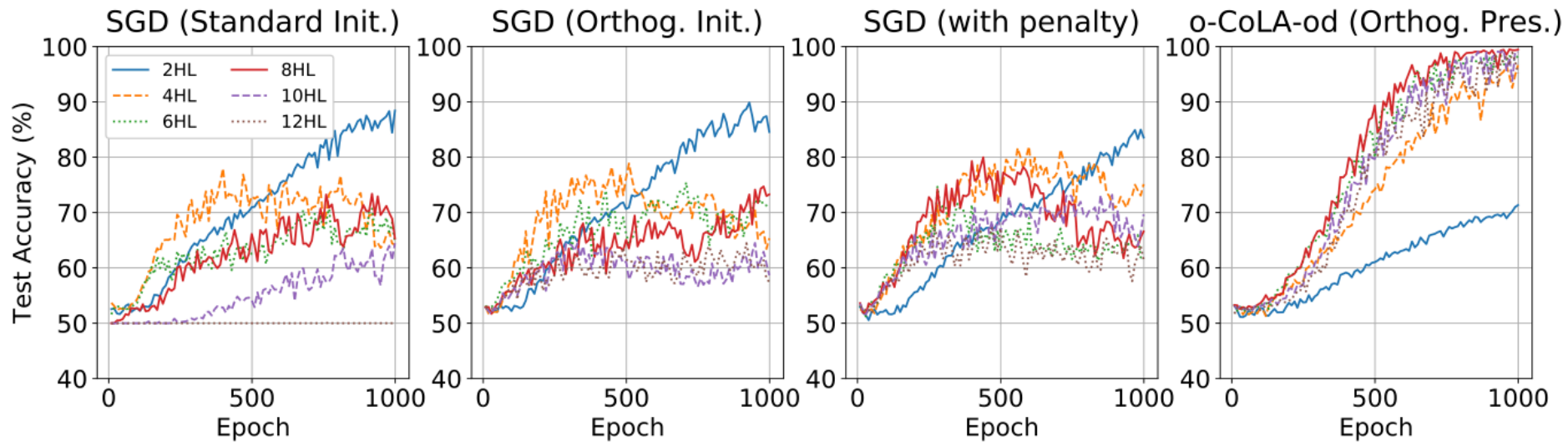


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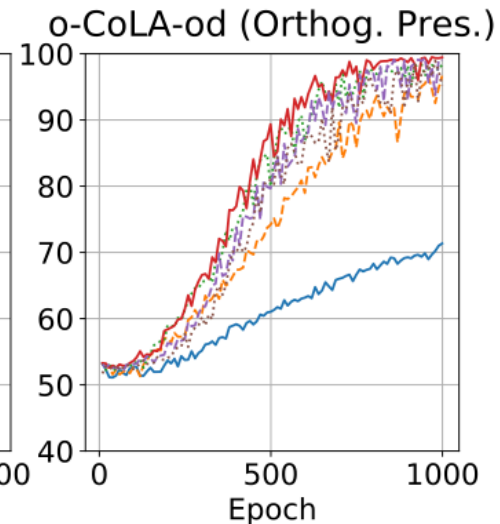
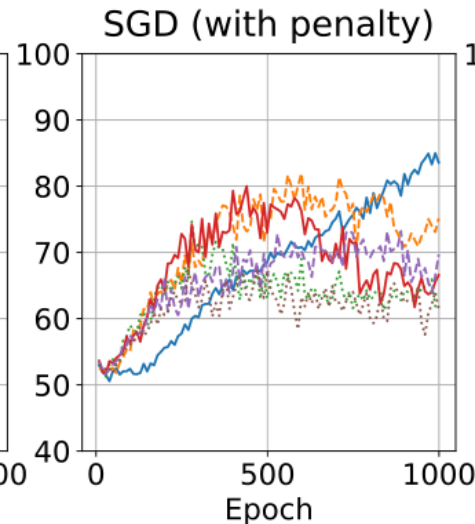
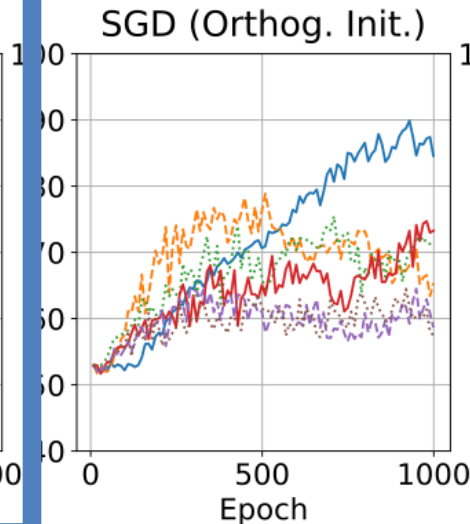
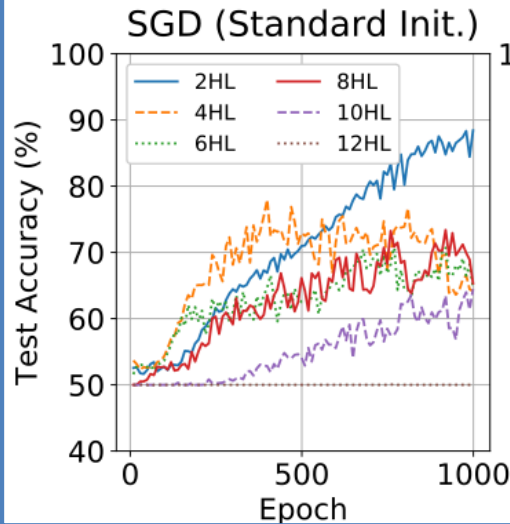


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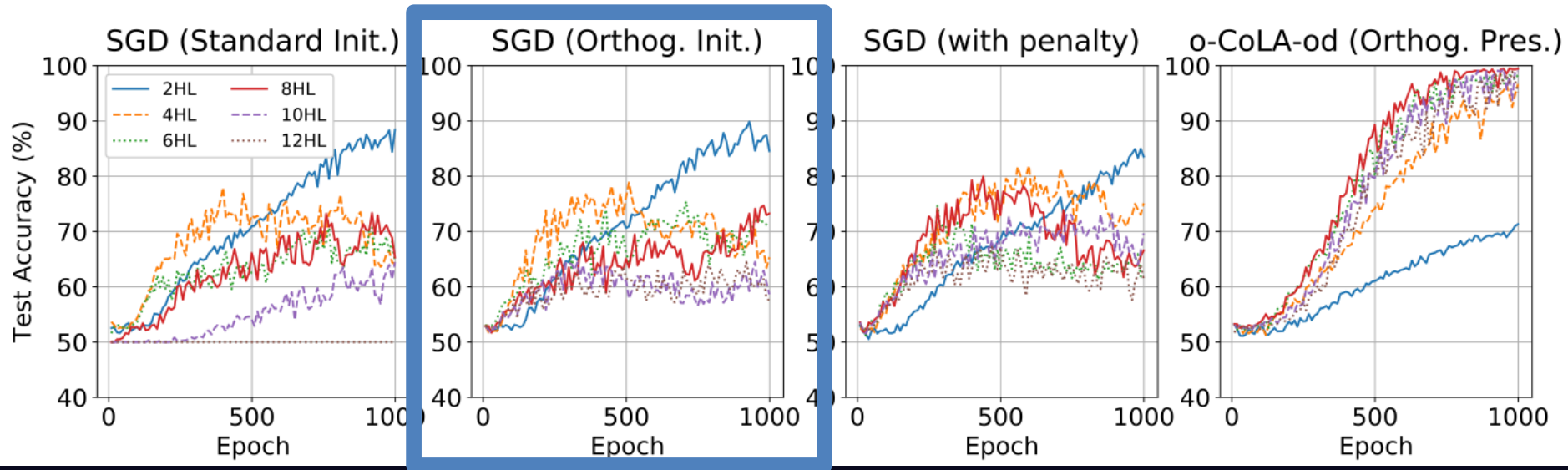
Orthogonal Constraints



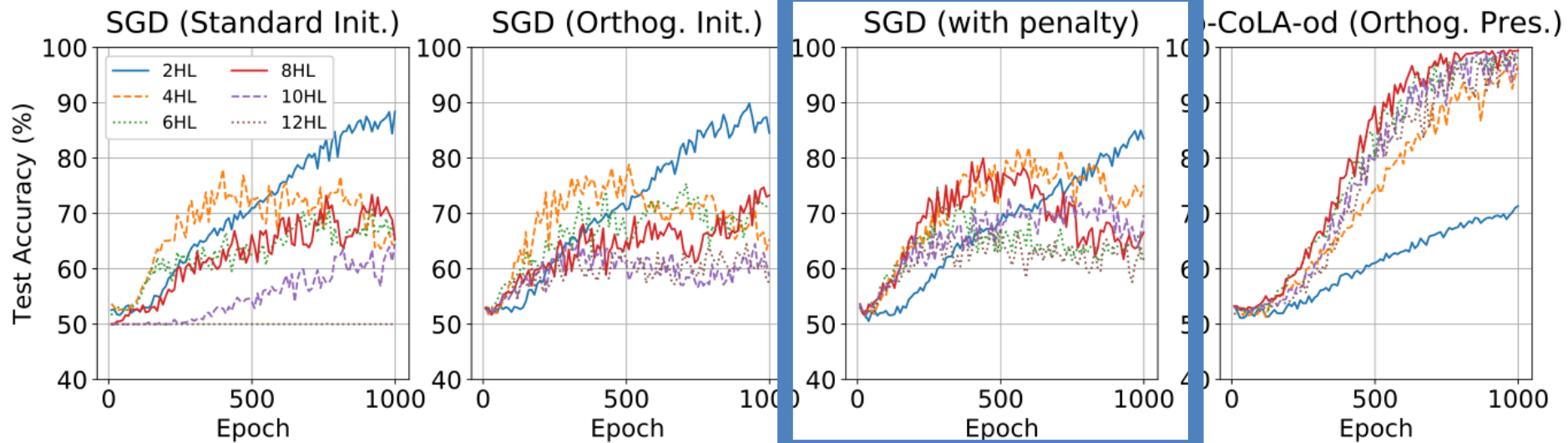
Orthogonal Constraints



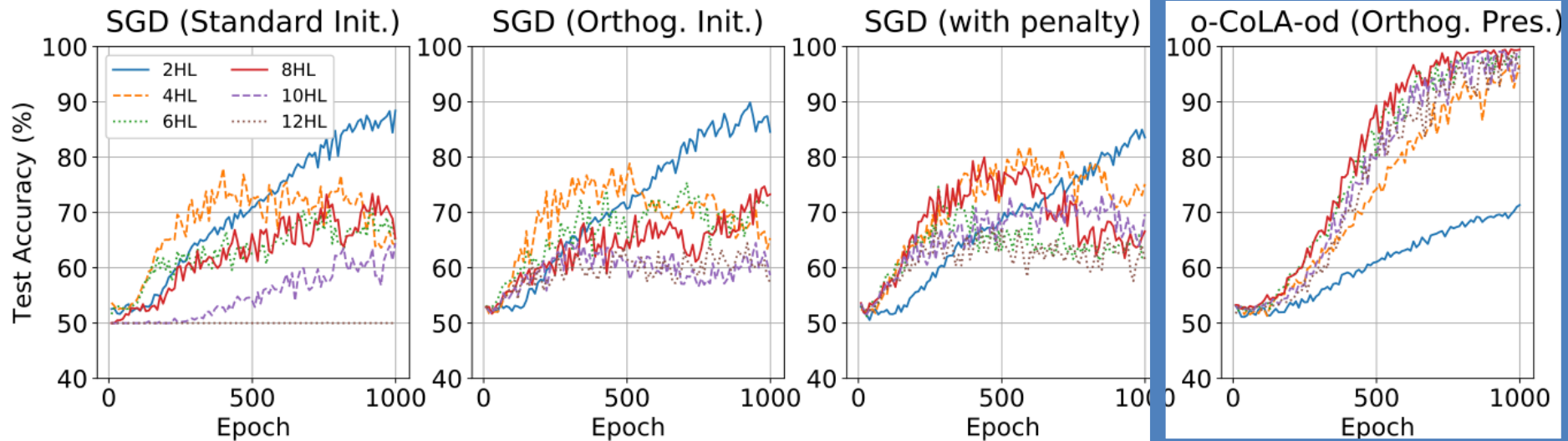
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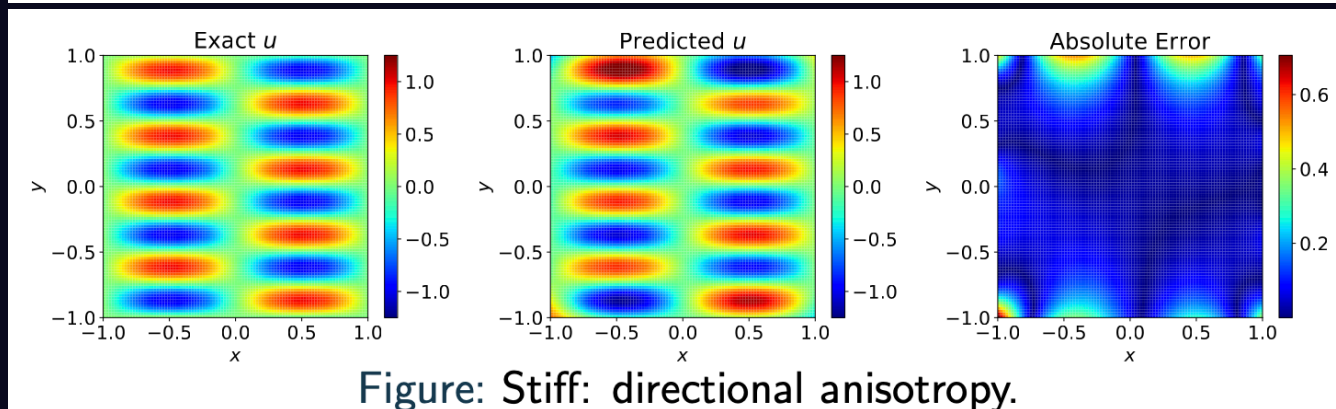
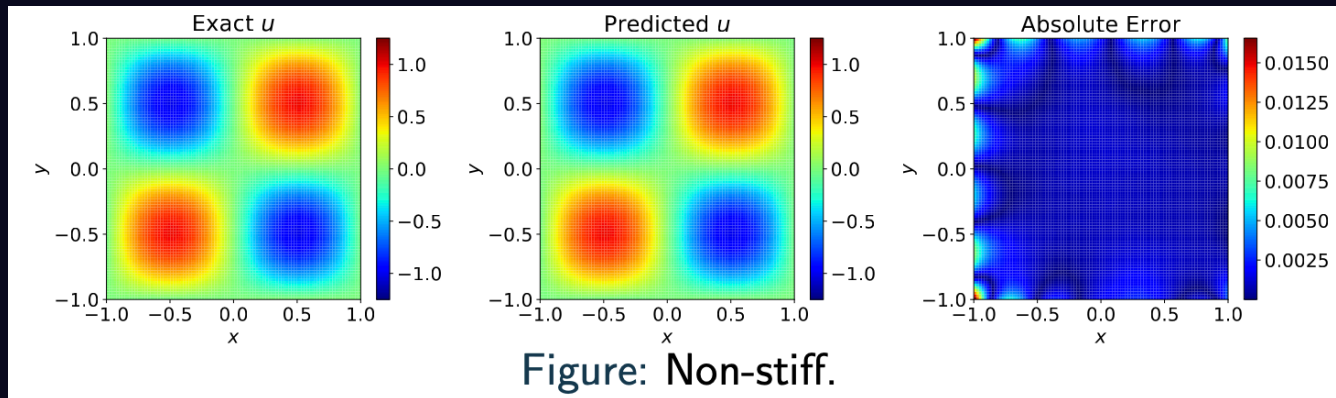
Physics-Informed Constraints for Neural Networks

- Soft constraints
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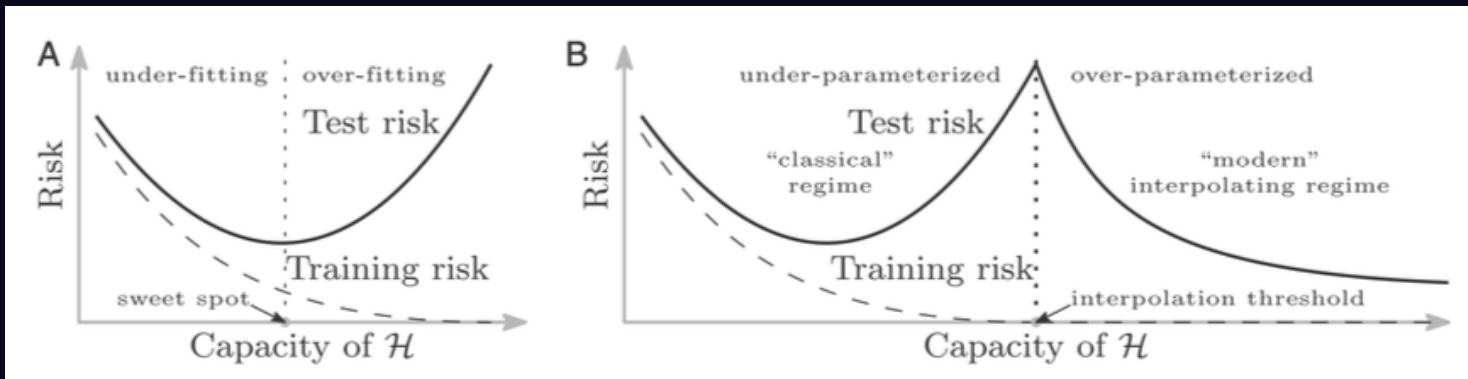
Example: 2D Helmholtz equation



Physics-Informed Constraints for Neural Networks

- Soft constraints
 - Physics-informed neural networks.
 - Widely used but suffer from various failure modes.
 - Working on ways to mitigate these.
- Hard constraints
 - Power flows on the electrical grid. Donti et al.
 - Physically consistent high-resolution climate data. Harder et al. 2022.

Role of Overparameterization



Belkin et al. 2019

Implicit Regularization

Neyshabur et al., 2015: In search of the real inductive bias: On the role of implicit regularization in deep learning.

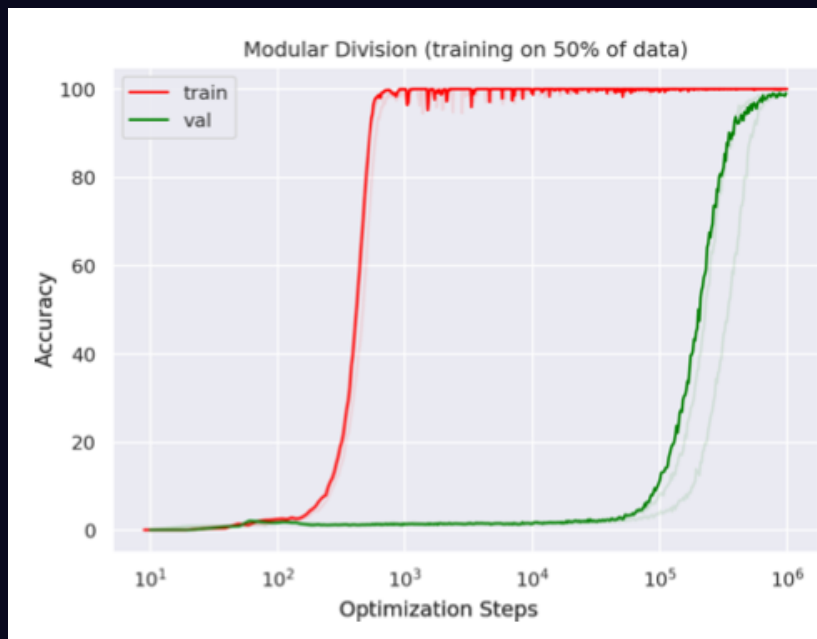
Imposed not by modifying the objective, but by the choice of training algorithm.

- small-batch vs. large-batch SGD training [Keskar et al., 2017]
- initial large learning rates [Li et al., 2019]
- early stopping [Yao et al., 2007]
- batch normalization [Ioffe and Szegedy, 2015]

More on optimization in the next lecture!

Grokking [Power et al., 2022]

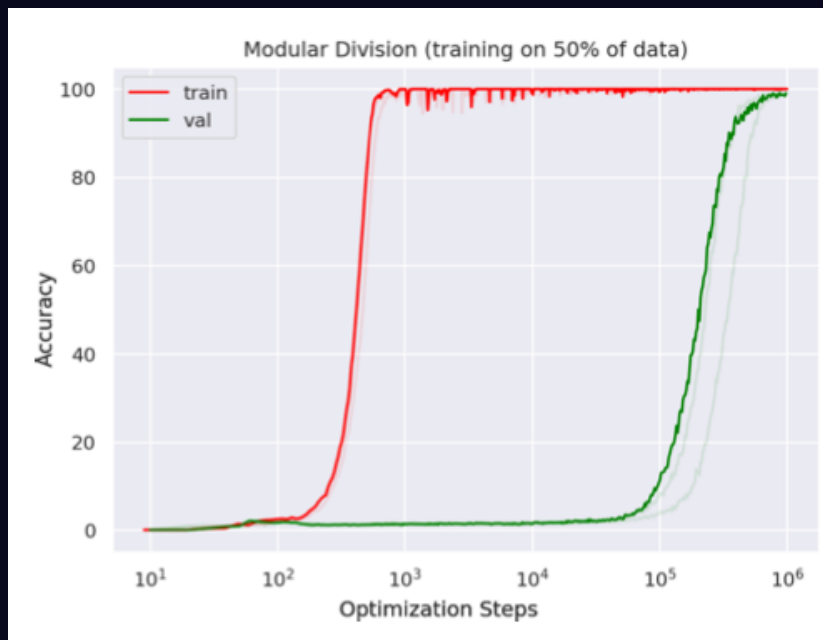
Remarkable delayed generalization pattern when training neural networks on algorithmic datasets (tasks like e.g. modular addition).



Long after the neural network had started to overfit on the training data, its performance on the validation data suddenly improved dramatically, reminiscent of a phase transition occurring.

Grokking [Power et al., 2022]

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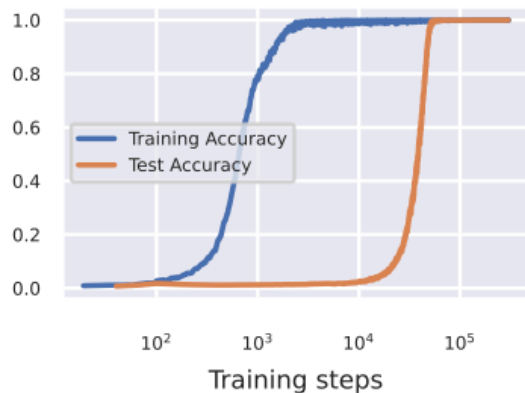


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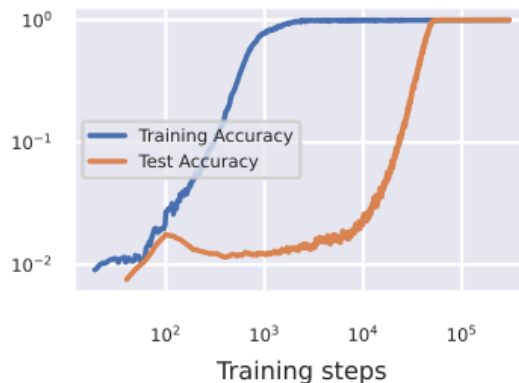
Omnigrok: Grokking Beyond Algorithmic Data, Liu et al, 2023.

Unifying grokking and double descent

Davies et al. 2023



(a) Typical grokking on a modular division task.



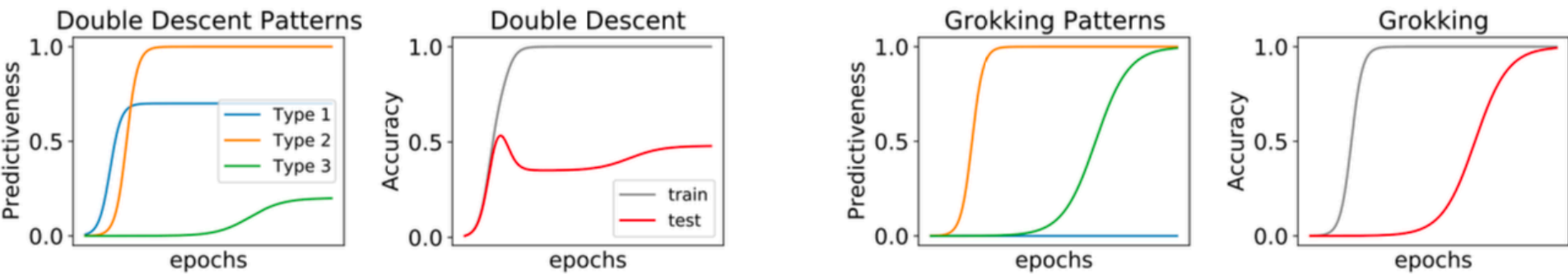
(b) Plotting log accuracy reveals a small double descent.



(c) The double descent is more pronounced in loss.

Unifying grokking and double descent

Davies et al. 2023



Type 3 pattern: slower to learn, better-generalizing.

“hypothesize that grokking and double descent can be understood as instances of the same learning dynamics within a framework of pattern learning speeds.”

Download Jupyter Notebook from:

tiffanyvlaar.github.io/VIASM

Access GPUs via Google Colab

In the tutorial there are:

- tasks
 - ➡ for reflection and engagement
 - ➡ for example: What is the effect of changing the amount of weight decay?
- challenges
 - ➡ code does NOT work, this is intentional